

(incoming)
 ① $j_N(u, r, z, \bar{z}) = \frac{1}{r^2} \delta(u) Y_{lm}(z, \bar{z})$, $j_r = j_z = 0$

(outgoing)
 $j_u(u, r, z, \bar{z}) = \frac{1}{r^2} \delta(u) Y_{lm}(z, \bar{z})$, $j_r = j_z = 0$

scatter at $u = v = 0$

$$\begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$$

(a) Find a solⁿ for F sourced by a null shockwave assuming

$$F_{\nu r} = \lim_{r \rightarrow \infty} [r^2 F_{\nu r}] = 0$$

and

$$\text{Support}(F_{\nu z}) = \{v=0\}$$

$$d * F = * j$$

$$\Rightarrow \nabla_\mu F^{\mu\sigma} = \frac{1}{\sqrt{-g}} \partial_\mu (\sqrt{-g} F^{\mu\sigma})$$

$$= \frac{1}{r^2} r^{2\bar{z}} \partial_\nu (r^2 r_{2\bar{z}} F^{\mu\sigma})$$

$$= \frac{1}{r^2} r^{2\bar{z}} [\partial_\nu (r^2 r_{2\bar{z}} F^{\nu\sigma}) + \partial_r (r^2 r_{2\bar{z}} F^{r\sigma}) + D_A (r^2 r_{2\bar{z}} F^{A\sigma})]$$

$$= \frac{1}{r^2} \partial_\nu (r^2 F^{\nu\sigma}) + \partial_r (r^2 r_{2\bar{z}} g^{r\alpha} g^{\sigma\beta} F_{\alpha\beta}) + D_A (r^2 r_{2\bar{z}} F^{A\sigma})$$

$$\sigma = r :$$

$$j_N e^2 = \frac{1}{r^2} \partial_\nu (r^2 F^{\nu r}) + D_A (r^2 r_{2\bar{z}} g^{A\bar{A}} F_{\bar{A}B} g^{rB})$$

$$= \frac{1}{r^2} \partial_\nu (r^2 F^{\nu r}) + D_z (F_{\bar{z}B} g^{rB}) + D_{\bar{z}} (F_{zB} g^{rB})$$

$$= \frac{1}{r^2} \partial_r (r^2 F_{rN}) + \underline{D_z (F_{\bar{z}r} + F_{\bar{z}N})} + \underline{D_{\bar{z}} (F_{zr} + F_{zN})}$$

$$\sigma = v :$$

$$\frac{1}{r^2} \partial_\nu (r^2 F^{\nu\nu}) + \partial_r (r^2 r_{2\bar{z}} g^{r\alpha} g^{\nu\beta} F_{\alpha\beta}) + D_A (r^2 r_{2\bar{z}} F^{A\nu})$$

$$= e^2 j^\nu = e^2 g^{\nu r} j_r = 0$$

$$\partial_r (r^2 r_{2\bar{z}} g^{r\alpha} g^{\nu\beta} F_{\alpha\beta}) + D_A (r^2 r_{2\bar{z}} g^{A\bar{A}} F_{\bar{A}B} g^{\nu B}) = 0$$

$$= \partial_r (r^2 r_{2\bar{z}} F_{\nu r}) + D_z (F_{\bar{z}B} g^{\nu B}) + D_{\bar{z}} (F_{zB} g^{\nu B}) = 0$$

$$= \partial_r (r^2 r_{2\bar{z}} F_{\nu r}) + D_z (F_{\bar{z}r}) + D_{\bar{z}} (F_{zr}) = 0$$

$$= -\partial_r (r^2 F_{rN}) + \dots = 0$$

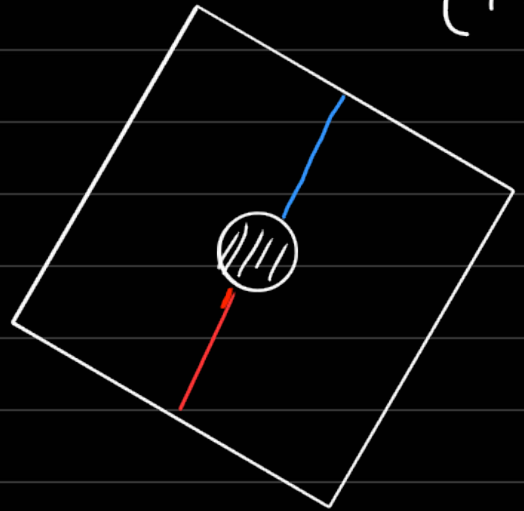
$$\Rightarrow \partial_r (r^2 F_{rN}) = \underline{r^{2\bar{z}} D_z (F_{\bar{z}r})} + \underline{r^{2\bar{z}} D_{\bar{z}} F_{zr}}$$

o.s.

$$\frac{1}{r^2} ((\partial_r + \partial_v)(r^2 F_{rN}) + r^{2\bar{z}} D_z F_{\bar{z}N} + r^{2\bar{z}} D_{\bar{z}} F_{zN}) = e^2 j_N$$

$$\text{or } \frac{1}{r^2} [(\partial_r + \partial_v)(r^2 F_{rN}) - r^{2\bar{z}} (\partial_z F_{N\bar{z}} + \partial_{\bar{z}} F_{Nz})] = e^2 j_N$$

$\Rightarrow$



$$\left[(\partial_r + \partial_v)(r^2 F_{rv}) - \gamma^{z\bar{z}} (\partial_z F_{r\bar{z}} + \partial_{\bar{z}} F_{rz}) \right] = e^2 \delta(v) \gamma_{\ell m}(z, \bar{z})$$

$$\partial_r(r^2 F_{rv}) + \gamma^{z\bar{z}} (\partial_z F_{r\bar{z}} + \partial_{\bar{z}} F_{rz}) = 0$$

$$\begin{aligned} \underline{Z} = v: \quad & \frac{\gamma^{z\bar{z}}}{r^2} \partial_v (r^2 F^{vz}) + \partial_r (r^2 \gamma_{z\bar{z}} g^{r\alpha} g^{z\beta} F_{\alpha\beta}) + D_A (r^2 \gamma_{z\bar{z}} F^{A\bar{z}}) = 0 \\ & = \partial_v (r^2 g^{v\alpha} g^{z\bar{z}} F_{\alpha\bar{z}}) + \partial_r (r^2 g^{r\alpha} F_{\alpha\bar{z}}) + \gamma^{z\bar{z}} D_{\bar{z}} (r^2 \gamma_{z\bar{z}} F^{\bar{z}z}) = 0 \\ & = \partial_v (F_{r\bar{z}} \gamma^{z\bar{z}}) + \partial_r ((F_{r\bar{z}} + F_{v\bar{z}}) \gamma^{z\bar{z}}) + \gamma^{z\bar{z}} D_{\bar{z}} (1) = 0 \end{aligned}$$

$$\Rightarrow \partial_v (F_{r\bar{z}}) + \partial_r (F_{r\bar{z}} + F_{v\bar{z}}) + D_{\bar{z}} (\gamma_{z\bar{z}} \frac{1}{r^2}) \gamma^{z\bar{z}} \gamma^{z\bar{z}} F_{z\bar{z}} = 0$$

$$\partial_v (F_{r\bar{z}}) + \partial_r (F_{r\bar{z}} + F_{v\bar{z}}) + D_{\bar{z}} (\gamma^{z\bar{z}} / r^2) F_{z\bar{z}} = 0$$

$$r^2 (\partial_v + \partial_r) F_{r\bar{z}} + r^2 \partial_r F_{v\bar{z}} + \partial_{\bar{z}} (\gamma^{z\bar{z}} F_{z\bar{z}}) = 0$$

Take $z \leftrightarrow \bar{z}$

$$\therefore r^2 (\partial_v + \partial_r) F_{rz} + r^2 \partial_r F_{vz} - \partial_z (\gamma^{z\bar{z}} F_{z\bar{z}}) = 0$$

Bianchi : $dF = 0 \quad \partial_{[\mu} F_{\nu\sigma]} = 0$

$$(\partial_\mu F_{\nu\sigma} - \partial_\nu F_{\mu\sigma} + \partial_\sigma F_{\mu\nu} - \partial_\sigma F_{\nu\mu} + \partial_\nu F_{\sigma\mu} - \partial_\mu F_{\sigma\nu})$$

$$* \partial_\mu F_{\nu\sigma} + \partial_\sigma F_{\mu\nu} + \partial_\nu F_{\sigma\mu} = 0$$

(i) (v, r, z)

$$\partial_v F_{rz} - \partial_z F_{rv} - \partial_r F_{vz} = 0$$

(ii) $(v, z, \bar{z})$

$$\partial_v F_{z\bar{z}} + \partial_{\bar{z}} F_{vz} - \partial_z F_{v\bar{z}} = 0$$

$$(iii) \partial_r F_{z\bar{z}} + \partial_{\bar{z}} F_{rz} - \partial_z F_{r\bar{z}} = 0 \quad (v, r, \bar{z})$$

* $S_{\text{sol}}^{\ell m}$ is dependent on a single $\gamma_{\ell m}$ and so each sector does not mix

$\Rightarrow$ Radial, Advanced Gauge $A_r = 0, A_v|_{\mathcal{I}^+} = 0$

$$F_{v\alpha}|_{\mathcal{I}^+} = \partial_v A_\alpha - \partial_\alpha A_v = 0 \quad \partial_v A_z = 0$$

$$A_{\bar{z}} = A_z(z, \bar{z}) / \bar{z}$$

$$F_{v\bar{z}}^{(0)} = 0 \text{ at } \mathcal{I}^+$$

$$\Rightarrow F_{vz} \propto \delta(v)$$

$$\partial A_{\bar{z}} - \bar{\partial} A_z \quad F_{r\bar{z}} = \partial_r A_{\bar{z}} - \partial A_v$$

$$F_{r\alpha} = \partial_r A_\alpha - \partial_\alpha A_r = \partial_r A_\alpha$$

$$F_{r\bar{r}} = \partial_r A_{\bar{r}} = 0 \Rightarrow A_{\bar{r}} = A_{\bar{r}}(r, z, \bar{z})$$

$$\text{so } F_{r\bar{z}} = F_{r\bar{z}}(r, z, \bar{z})$$

$$(i) \partial_r F_{r\bar{z}} - \partial_r F_{r\bar{z}} = 0$$

$$\partial_r F_{r\bar{z}} = 0 \Rightarrow F_{r\bar{z}} = F_{r\bar{z}}(r, z, \bar{z})$$

$$\partial_r A_{\bar{z}} = 0$$

$$F_{r\bar{z}} = 0 = F_{r\bar{r}}$$

$$(ii) \partial_r F_{z\bar{z}} = 0 \Rightarrow F_{z\bar{z}} = F_{z\bar{z}}(r, z, \bar{z})$$

$$\Rightarrow \partial_z (r^{z\bar{z}} F_{z\bar{z}}) = 0$$

$$(\partial_z r^{z\bar{z}}) F_{z\bar{z}} + r^{z\bar{z}} \partial_z F_{z\bar{z}} = 0$$

$$(1 + z\bar{z}) \bar{z} F_{z\bar{z}} + r^{z\bar{z}} \partial_z F_{z\bar{z}} = 0$$

$$\int dz \left(\frac{z}{1+z\bar{z}} \right) F_{z\bar{z}} + \partial_z F_{z\bar{z}} = 0$$

$$- F_{z\bar{z}} \Big|_{r^1} = \int dz \left(\frac{z}{1+z\bar{z}} \right) F_{z\bar{z}}$$

$$\partial \ln(1+z\bar{z}) F_{z\bar{z}} dz = - F_{z\bar{z}}$$

• Take $\bar{\partial}$ of other equation as this is analytic to finally have $F_{z\bar{z}} = -F_{z\bar{z}} \Rightarrow F_{z\bar{z}} = 0$

$$F_{z\bar{z}} = 0$$

This leaves

$$-r^{z\bar{z}} (\partial F_{r\bar{z}} + \bar{\partial} F_{r\bar{z}}) = e^2 \delta(r) \gamma_{em}$$

$$\partial F_{r\bar{z}} = \bar{\partial} F_{r\bar{z}}$$

$$\Rightarrow -2r^{z\bar{z}} \bar{\partial} F_{r\bar{z}} = e^2 \delta(r) \gamma_{em}$$

$$\begin{cases} \bar{\partial} F_{r\bar{z}} = -\frac{1}{2} r^{z\bar{z}} e^2 \delta(r) \gamma_{em} \\ \partial F_{r\bar{z}} = -\frac{1}{2} r^{z\bar{z}} e^2 \delta(r) \gamma_{em} \end{cases}$$

$$F_{r\bar{z}} = -\frac{1}{2} e^2 \delta(r) M_{em}(z, \bar{z})$$

$$\bar{\partial} F_{r\bar{z}} = -\frac{1}{2} e^2 \delta(r) \bar{\partial} M_{em}$$

$$\bar{\partial} M_{em} = r^{z\bar{z}} \gamma_{em}$$

$$\text{By symmetry, let } M_{em} = \partial \mathcal{U}_{em}$$

$$F_{r\bar{z}} = -\frac{1}{2} e^2 \delta(r) \partial \mathcal{U}_{em}$$

$$\Rightarrow \partial \bar{\partial} \mathcal{U}_{em} = r^{z\bar{z}} \gamma_{em}$$

$$(00): \partial \bar{\partial} \mathcal{U}_{00} = \frac{z}{(1+z\bar{z})^2} \gamma_{00}$$

$$\partial \bar{\partial} \ln(1+z\bar{z}) = \partial \left(\frac{z}{1+z\bar{z}} \right) = \frac{1}{1+z\bar{z}} - \frac{z\bar{z}}{(1+z\bar{z})^2} = \frac{z}{(1+z\bar{z})^2}$$

$$\Rightarrow \partial \bar{\partial} \chi_{00} = 2 \partial \bar{\partial} \ln(1+z\bar{z}) \gamma_{00}$$

$$\text{or } \chi_{00} = 2 \ln(1+z\bar{z}) \gamma_{00} + f(r)$$

$$(F_{rz})_{00} = -e^2 \delta(r) \partial \chi_{00} \quad | \quad \chi_{00} = \frac{r_{00}}{2}$$

or

$$(F_{rz})_{00} = -e^2 \delta(r) (\partial \ln(1+z\bar{z})) \gamma_{00}$$

$(lm) \neq 0$:

$$\partial \bar{\partial} \chi_{lm} = \gamma_{z\bar{z}} \gamma_{lm} = 2 \partial \bar{\partial} \chi_{lm} = \gamma_{z\bar{z}} \gamma_{lm}$$

$$D^2 f = D^A D_A f = (D^{\bar{z}} D_{\bar{z}} + D^{\bar{z}} D_{\bar{z}}) f$$

$$\Rightarrow D^2 = 2 \gamma^{z\bar{z}} \partial_{\bar{z}} \partial_{\bar{z}}$$

$$D^2 \gamma_{lm} = -l(l+1) \gamma_{lm}$$

$$\Rightarrow 2 \partial \bar{\partial} \chi_{lm} = \gamma_{z\bar{z}} \left(\frac{-1}{l(l+1)} \right) 2 \gamma^{z\bar{z}} \partial \bar{\partial} \chi_{lm}$$

$$\chi_{lm} = - \frac{\gamma_{lm}}{l(l+1)} \quad l \neq 0$$

$$F_{rz} = -e^2 \delta(r) \partial \chi_{lm}$$

$$\chi_{lm} = \begin{cases} \partial \ln(1+z\bar{z}) \gamma_{00}, & l=0 \\ -\frac{1}{l(l+1)} \gamma_{lm}, & l \neq 0 \end{cases}$$

• $A_r = 0$ Gauge?

$$F_{rz} = \partial_r A_z = -e^2 \delta(r) \partial \chi_{lm}$$

$$A_z(0^+, z, \bar{z}) = -e^2 \Theta(r) \partial \chi_{lm}, \quad A_r = A_r = 0$$

$$F_{uz} = -e^2 \delta(u) \partial \chi'_{lm}$$

$$(b) \quad A_z^{(0)}|_{\bar{z}^-} = 0 \quad | \quad A_z^{(0)}|_{\bar{z}^+} = A_z^{(0)}|_{\bar{z}^-}$$

$$A_z^{(0)}(u, z, \bar{z}) = -e^2 \Theta(u) \partial \chi_{lm}$$

$$\Rightarrow A_z^{(0)} = -e^2 \partial \chi_{lm}(z, \bar{z})$$

$$\Rightarrow F_{uz} = \partial_u A_z = -e^2 \delta(u) \partial \chi_{lm}$$

$$A_z(u, z, \bar{z}) = -e^2 \Theta(u) \partial \chi_{lm} - e^2 f(z, \bar{z}) \partial \chi_{lm}$$

$$\lim_{u \rightarrow \infty} A_z = A_z|_{\bar{z}^+} = -e^2 \partial \chi_{lm} \quad \text{so } f = 1$$

$$A_z(u, z, \bar{z}) = -e^2 \Theta(u) \partial \chi_{lm} - e^2 \partial \chi'_{lm}$$

② Lorentz-Gauge $\nabla^\mu A_\mu = 0$, Pure Y.M.
 * Use residual Gauge freedom to impose $A_u^{(1)} = 0$

$$\nabla_\mu F^{\mu\nu} = 0 \Rightarrow \partial_\mu (\sqrt{-g} F^{\mu\nu}) = 0$$

Boundary/Initial Conditions: $A_z^{(0)}(u, z, \bar{z}) \stackrel{lim}{\rightarrow r \rightarrow \infty} A_2(u, r, z, \bar{z})$
 $(u, r, z, \bar{z})$ $ds^2 = -du^2 - 2dzdr + \dots$

Maxwell: $g_{\alpha\beta} \sim \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ $g^{\alpha\beta} \sim \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$$\partial_u (r^2 \gamma_{z\bar{z}} F^{u\sigma}) + \partial_r (r^2 \gamma_{z\bar{z}} F^{r\sigma}) + \partial_A (r^2 \gamma_{z\bar{z}} F^{A\sigma}) = 0$$

$$\partial_u (r^2 \gamma_{z\bar{z}} g^{u\alpha} g^{\sigma\beta} F_{\alpha\beta}) + \partial_r (r^2 \gamma_{z\bar{z}} g^{r\alpha} g^{\sigma\beta} F_{\alpha\beta}) + \partial_A (r^2 \gamma_{z\bar{z}} g^{A\alpha} F_{\alpha\beta} g^{\sigma\beta}) = 0$$

$$\left\{ \begin{aligned} \partial_u (r^2 \gamma_{z\bar{z}} (-1) g^{\sigma B} F_{rB}) + \partial_r (r^2 \gamma_{z\bar{z}} g^{\sigma B} (-F_{uB} + F_{rB})) \\ \partial_A (F_{AB} g^{\sigma B}) = 0 \end{aligned} \right\}$$

u-term:

$$-\partial_u (r^2 \gamma_{z\bar{z}} g^{uB} F_{rB}) + \partial_r (r^2 \gamma_{z\bar{z}} g^{uB} (-F_{uB} + F_{rB})) + \partial_A (F_{AB} g^{uB}) = 0$$

$$\partial_r (r^2 \gamma_{z\bar{z}} (F_{ur})) - \partial_z (F_{\bar{z}r}) - \partial_{\bar{z}} (F_{zr})$$

or

$$-\partial_r (r^2 F_{ru}) + \gamma^{z\bar{z}} (\partial_z F_{r\bar{z}} + \partial_{\bar{z}} F_{rz}) = 0$$

r-term:

$$\partial_u (r^2 \gamma_{z\bar{z}} F_{ru}) + \partial_r (r^2 \gamma_{z\bar{z}} g^{rB} (-F_{uB} + F_{rB})) + \partial_A (F_{AB} g^{rB}) = 0$$

$$= \partial_u (r^2 \gamma_{z\bar{z}} F_{ru}) + \partial_r (r^2 \gamma_{z\bar{z}} (F_{ru} - F_{ur})) + \partial_A (F_{AB} g^{rB}) = 0$$

$$= \partial_u (r^2 \gamma_{z\bar{z}} F_{ru}) + \partial_z (F_{\bar{z}r} - F_{\bar{z}u}) + \partial_{\bar{z}} (F_{zr} - F_{zu}) = 0$$

$$\partial_u (r^2 F_{ru}) - \gamma^{z\bar{z}} (\partial_z F_{r\bar{z}} + \partial_{\bar{z}} F_{rz}) - \gamma^{z\bar{z}} (\partial_z F_{\bar{z}u} + \partial_{\bar{z}} F_{zu})$$

$$\Rightarrow (\partial_u - \partial_r) (r^2 F_{ru}) + \gamma^{z\bar{z}} (\partial_{\bar{z}} F_{ru} + \bar{\partial} F_{ru}) = 0$$

(from adding u-term)

z-term:

$$-\partial_u (F_{r\bar{z}}) + \partial_r (-F_{u\bar{z}} + F_{r\bar{z}}) + \partial_{\bar{z}} (F_{z\bar{z}} \frac{1}{r^2} \gamma^{z\bar{z}}) = 0$$

$$r^2 (\partial_r - \partial_u) (F_{r\bar{z}}) - r^2 \partial_r F_{uz} - \partial (F_{z\bar{z}} \gamma^{z\bar{z}}) = 0$$

or

$$r^2 (\partial_u - \partial_r) (F_{r\bar{z}}) + r^2 \partial_r F_{uz} + \partial_z (F_{z\bar{z}} \gamma^{z\bar{z}}) = 0$$

Lorentz Condition:

$$\nabla_\mu A^\mu = \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} A^\mu) = 0 = \partial_\mu (r^2 \gamma_{z\bar{z}} A^\mu)$$

$$0 = \partial_u (r^2 \gamma_{z\bar{z}} g^{u\sigma} A_\sigma) + \partial_r (r^2 \gamma_{z\bar{z}} g^{r\sigma} A_\sigma) + \partial_z (r^2 \gamma_{t\bar{z}} g^{t\sigma} A_\sigma)$$

$$0 = \partial_u (r^2 \gamma_{z\bar{z}} (-1) A_r) + \partial_r (r^2 \gamma_{z\bar{z}} (A_r - A_u)) + \partial_z (A_{\bar{z}}) + \partial_{\bar{z}} A_z$$

$$\text{or}$$

$$\boxed{(\partial_r - \partial_u)(r^2 A_r) - \partial_r (r^2 A_u) + \gamma^{z\bar{z}} (\partial_z A_{\bar{z}} + \partial_{\bar{z}} A_z) = 0}$$

Next step: Expand each set of Maxwell's eq in A and impose Gauge to each:

u-term:

$$-\partial_r (r^2 F_{ru}) + \gamma^{z\bar{z}} (\partial_z F_{r\bar{z}} + \partial_{\bar{z}} F_{rz}) = 0$$

$$= -\partial_r (r^2 (\partial_r A_u - \partial_u A_r)) + \gamma^{z\bar{z}} [(\partial_z \partial_r A_{\bar{z}} - \partial_{\bar{z}} \partial_r A_r) + (\partial_{\bar{z}} \partial_r A_z - \partial_{\bar{z}} \partial_r A_r)]$$

$$-\partial_r (r^2 (\partial_r A_u - \partial_u A_r)) - 2 \gamma^{z\bar{z}} \partial_{\bar{z}} \partial_r A_r + \partial_r [\gamma^{z\bar{z}} (\partial_z A_{\bar{z}} + \partial_{\bar{z}} A_z)]$$

Plugging in $\gamma^{z\bar{z}} (\partial_z A_{\bar{z}} + \partial_{\bar{z}} A_z)$

$$= (\partial_u - \partial_r)(r^2 A_r) + \partial_r (r^2 A_u)$$

$$-\partial_r (r^2 (\partial_r A_u - \partial_u A_r)) - 2 \gamma^{z\bar{z}} \partial_{\bar{z}} \partial_r A_r + \partial_r ((\partial_u - \partial_r)(r^2 A_r) + \partial_r (r^2 A_u))$$

$$-\partial_r (r^2 \partial_r A_u) + \partial_r (r^2 \partial_u A_r) + \partial_r (r^2 \partial_u A_r) - \partial_r^2 (r^2 A_r) + \partial_r (r^2 A_u)$$

$$2 \partial_r (r^2 \partial_u A_r) - 2 r \partial_r A_u - r^2 \partial_r^2 A_u - \partial_r^2 (r^2 A_r) + \partial_r (r^2 A_u)$$

$$- 2 \gamma^{z\bar{z}} \partial_{\bar{z}} \partial_r A_r$$

$$- 2 r \partial_r A_u - r^2 \partial_r^2 A_u + \partial_r (r^2 A_u)$$

$$= - 2 \partial_r (r A_u) + 2 A_u$$

* Three Branches + Gauge:

$$\left\{ \begin{aligned} - 2 r \partial_r (r \partial_u A_u) + \partial_r (r^2 \partial_r A_u) + 2 \gamma^{z\bar{z}} \partial_z \partial_{\bar{z}} A_u &= 0 \\ - 2 \partial_r (r A_u) - 2 \partial_u \partial_r (r^2 A_r) + \partial_r^2 (r^2 A_r) + 2 \gamma^{z\bar{z}} \partial_z \partial_{\bar{z}} A_r &= 0 \\ - 2 r^2 \partial_u \partial_r A_z + r^2 \partial_r^2 A_z + 2 r \partial_r (A_r - A_u) + 2 \partial_z (\gamma^{z\bar{z}} \partial_{\bar{z}} A_z) &= 0 \end{aligned} \right.$$

Gauge field expansions: $A_\mu^{(n)} = \frac{1}{r^n} A_\mu(z, r, \bar{z}, \bar{r})$

* Gauge $A_u^{(n)} = 0$

$$A_z(z, r, \bar{z}, \bar{r}) = \sum_{n=0}^{\infty} \frac{1}{r^n} A_z^{(n)}$$

$$A_r = \sum_{n=1}^{\infty} \frac{1}{r^n} A_r^{(n)}, \quad A_{\bar{z}} = \sum_{n=2}^{\infty} \frac{1}{r^n} A_{\bar{z}}^{(n)}$$

→ Comes from the Lorentz Gauge lemma on indexed set

of residual gauge transformations: $\square \mathcal{E} = 0$

$\mathcal{E}_r \sim \mathcal{O}(\frac{1}{r})|_{\mathcal{I}^+} \Rightarrow$ can choose $A_r^{(0)} = 0$

Lorentz Gauge: $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$$\nabla^\mu A_\mu = \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} A^\mu)$$

$$= \frac{1}{r^2} \gamma^{2\bar{2}} \partial_\mu (r^2 \gamma_{2\bar{2}} A^\mu)$$

$$= \frac{1}{r^2} \gamma^{2\bar{2}} \left[\partial_u (r^2 \gamma_{2\bar{2}} A^u) + \partial_r (r^2 \gamma_{2\bar{2}} A^r) + \partial_A (r^2 \gamma_{2\bar{2}} A^A) \right]$$

$$= \frac{1}{r^2} \partial_u (g^{ur} A_r) + \frac{1}{r^2} \partial_r (r^2 (g^{rr} A_r + g^{rA} A_A)) + \dots$$

$$(\partial_r - \partial_u)(r^2 A_r) - \partial_r (r^2 A_u) + r^{2\bar{2}} \partial_A (A_{\bar{A}}) = 0$$

$\Rightarrow$

$$(\partial_r - \partial_u) \left(r^2 \sum_{n=0}^{\infty} \frac{1}{r^n} A_r^{(n)} \right) - \partial_r \left(r^2 \sum_{n=2}^{\infty} \frac{1}{r^n} A_u^{(n)} \right) + r^{2\bar{2}} \partial_A (A_{\bar{A}}) = 0$$

To lowest order, we have:

$$(\partial_r - \partial_u) \left(A_r^{(2)} \right) - \partial_r (A_u^{(2)}) + r^{2\bar{2}} \partial_2 A_{\bar{2}}^{(0)} + \partial_{\bar{2}} A_2^{(0)}$$

$$\partial_u A_r^{(2)} = r^{2\bar{2}} (\partial_2 A_{\bar{2}}^{(0)} + \partial_{\bar{2}} A_2^{(0)})$$

1st order:

$$\begin{aligned} & (\partial_r - \partial_u) \left(\frac{1}{r} A_r^{(3)} \right) - \partial_r \left(\frac{1}{r} A_u^{(3)} \right) + r^{2\bar{2}} \frac{1}{r} (\partial_2 A_{\bar{2}}^{(1)} + \partial_{\bar{2}} A_2^{(1)}) = 0 \\ & - \frac{1}{r} \partial_u A_r^{(3)} + \partial_r \left(\frac{1}{r} (A_r^{(3)} - A_u^{(3)}) \right) + r^{2\bar{2}} \frac{1}{r} (\partial_2 A_{\bar{2}}^{(1)} + \partial_{\bar{2}} A_2^{(1)}) = 0 \\ & - \frac{1}{r} \partial_u A_r^{(3)} - \frac{1}{r^2} (A_r^{(3)} - A_u^{(3)}) + \frac{1}{r} r^{2\bar{2}} (\partial_2 A_{\bar{2}}^{(1)} + \partial_{\bar{2}} A_2^{(1)}) = 0 \end{aligned}$$

$\Rightarrow$ Evaluated at $\mathcal{I}^+$:

$$\partial_u A_r^{(3)} = r^{2\bar{2}} (\partial_2 A_{\bar{2}}^{(1)} + \partial_{\bar{2}} A_2^{(1)})$$

2nd order:

$$\begin{aligned} & (\partial_r - \partial_u) \left(\frac{1}{r^2} A_r^{(4)} \right) - \partial_r \left(\frac{1}{r^2} A_u^{(4)} \right) + \frac{1}{r^2} r^{2\bar{2}} (\partial_2 A_{\bar{2}}^{(2)} + \partial_{\bar{2}} A_2^{(2)}) \\ & - \frac{2}{r^3} A_r^{(4)} - \frac{1}{r^2} \partial_u A_r^{(4)} + \frac{2}{r^3} A_u^{(4)} + \frac{1}{r^2} \end{aligned}$$

$$(\partial_r - \partial_u) \left(\sum_{n=2}^{\infty} r^{2-n} A_r^{(n)} \right) - \partial_r \left(\sum_{n=2}^{\infty} r^{2-n} A_u^{(n)} \right) + r^{2\bar{2}} \partial \dots$$

$m = n - 2$

$$(\partial_r - \partial_u) \left(\sum_{m=0}^{\infty} r^{-m} A_r^{(m+2)} \right) - \partial_r \left(\sum_{m=0}^{\infty} r^{-m} A_u^{(m+2)} \right) + r^{2\bar{2}} \partial \dots$$

$$\sum_{m=0}^{\infty} \left[-m r^{-m-1} A_r^{(m+2)} - r^{-m} \partial_u A_r^{(m+2)} + m r^{-m-1} A_u^{(m+2)} + r^{-m} \partial A_{\bar{2}}^{(m)} + \dots \right]$$

$$\sum_{l=0}^{\infty} l = m + 1$$

$$\sum_{l=0}^{\infty} \left[-(l-1) r^{-l} A_r^{(l)} + (l-1) r^{-l} A_z^{(l)} - r^{-l} \partial_z A_r^{(l)} + r^{-l} \gamma^{z\bar{z}} (\partial_z A_{\bar{z}}^{(l)} + \partial_{\bar{z}} A_z^{(l)}) \right] = 0$$

$\Rightarrow n \geq 1$:

$$(n-1) (A_z^{(n+1)} - A_r^{(n+1)}) + \gamma^{z\bar{z}} (\partial_z A_{\bar{z}}^{(n)} + \partial_{\bar{z}} A_z^{(n)}) = \partial_z A_r^{(n+2)}$$

$\Rightarrow$ Going back to Bianchi:

$$-2r \partial_r (r \partial_z A_n) + \partial_r (r^2 \partial_r A_n) + 2\gamma^{z\bar{z}} \partial_z \partial_{\bar{z}} A_n = 0$$

leading order term:

$$-2r \partial_r \left(\frac{1}{r} \partial_z A_z^{(2)} \right) + \partial_r \left(r^2 \left(-\frac{2}{r^3} A_z^{(2)} \right) \right) + 2\gamma^{z\bar{z}} \partial_z \partial_{\bar{z}} \frac{1}{r^2} A_z^{(2)} = 0$$

$$2r \left(\frac{1}{r^2} \partial_z A_n^{(2)} \right) + 2 \frac{1}{r^2} A_n^{(2)} + 2 \frac{1}{r^2} \gamma^{z\bar{z}} (\partial_z \partial_{\bar{z}} A_z^{(2)}) = 0$$

$$2 \partial_z A_n^{(2)} + \frac{1}{r} (\dots) = 0$$

So at $\mathcal{O}^+$: $\partial_z A_n^{(2)} = 0$

$n \geq 1$:

$$-2r \partial_r \left(\sum_{n=2}^{\infty} r^{-n} \partial_z A_z^{(n)} \right) + \partial_r \left(\sum_{n=2}^{\infty} r^2 (-n) r^{-n-1} A_z^{(n)} \right) + 2\gamma^{z\bar{z}} \sum_{n=2}^{\infty} r^{-n} \partial_z \partial_{\bar{z}} A_z^{(n)} = 0$$

$$-2r \sum_{n=2}^{\infty} (1-n) r^{-n} \partial_z A_z^{(n)} + \sum_{n=2}^{\infty} (1-n) r^{-n} (-n) A_z^{(n)} + 2\gamma^{z\bar{z}} \sum_{n=2}^{\infty} r^{-n} A_z^{(n)} = 0$$

$$-2 \sum_{n=2}^{\infty} r^{-n+1} \quad -n+1 = -n$$

$$-2 \sum_{m=1}^{\infty} r^{-m} (-m) \partial_z A_z^{(m+1)} + \dots$$

$$-2 \sum_{m=2}^{\infty} r^{-m} (-m) \partial_z A_z^{(m+1)} + \dots$$

$$2n \partial_z A_z^{(n+1)} - n(n-1) A_z^{(n)} + 2\gamma^{z\bar{z}} A_z^{(n)} = 0$$

$$\Rightarrow \partial_z A_z^{(n+1)} = -\frac{1}{2n} [n(n-1) + 2\gamma^{z\bar{z}}] A_z^{(n)} = 0 \quad n \geq 1$$

So all of A_n can be determined up to n1.

constants:

$$\partial_z A_z^{(2)} = 0 \Rightarrow A_z^{(2)} = f(z, \bar{z}) + C$$

$$\partial_z A_z^{(3)} = -\frac{1}{2} [1 + 2\gamma^{z\bar{z}}] f(z, \bar{z}) + \bar{z}$$

$$\Rightarrow A_n^{(3)} = \left[-2 + \frac{1}{2} \gamma^{2\bar{2}} \right] (f(z, \bar{z}) + c) u + g(z, \bar{z})$$

$$\partial_u A_n = \sum_{m=2}^{\infty} u^{(m)} (\partial_u A_m^{(n)}) = (n-2) + \sum_{m=2}^{\infty} u^m - \frac{1}{2n} [n(n-1) + 2\gamma^{2\bar{2}} \partial_{\bar{z}}] A_n^{(n-1)}$$

* Second Bianchi:

$$\begin{aligned} & -2\partial_r(rA_n) - 2\partial_u\partial_r(r^2A_r) + \partial_r^2(r^2A_r) + 2\gamma^{2\bar{2}}\partial_z\partial_{\bar{z}}A = 0 \\ & -2\sum_{n=2}^{\infty} (1-n)r^{-n}A_n^{(n)} - 2(2-n)r^{-n+1}\partial_uA_r^{(n)} \\ & + \sum_{n=2}^{\infty} (2-n)(1-n)r^{-n}A_r^{(n)} + 2\gamma^{2\bar{2}}\partial_z\partial_{\bar{z}}\sum_{n=2}^{\infty} r^{-n}A_r^{(n)} = 0 \\ \Rightarrow & \sum_{n=2}^{\infty} 2(1-n)r^{-n}A_n^{(n)} + \sum_{n=2}^{\infty} 2(n-1)r^{-n}\partial_uA_r^{(n+1)} + \sum_{n=2}^{\infty} (n-1)(n-2)A_r^{(n)} \\ & + 2(n-1)A_n^{(n)} + 2(n-1)\partial_uA_r^{(n+1)} + (n-1)(n-2)A_r^{(n)} + 2\gamma^{2\bar{2}}\partial_z\partial_{\bar{z}}A_r^{(n)} \\ & = 0 \end{aligned}$$

$$\partial_u A_r^{(n+1)} = -A_n^{(n)} - \left[\frac{1}{2}(n-2) + \frac{\gamma^{2\bar{2}}}{(n-1)} \partial_z \partial_{\bar{z}} \right] A_r^{(n)} = 0$$

$n=2$:

$$\partial_u A_r^{(3)} = -A_n^{(2)} - \gamma^{2\bar{2}} \partial_z \partial_{\bar{z}} A_r^{(2)} = 0 \dots$$

Third Equation:

$$\partial_u A_2^{(n)} = -\partial_z(\gamma^{2\bar{2}} \partial_{\bar{z}} A_2^{(n)}) \quad n \geq 0$$

$$\partial_u A_2^{(n+1)} = \left[\frac{1}{n+1} \right] \left[-\partial_z A_r^{(n+1)} + \partial_z A_n^{(n)} - \frac{1}{2} n(n+1) A_2^{(n)} - \partial_z(\gamma^{2\bar{2}} \partial_{\bar{z}} A_2^{(n)}) \right]$$

$$\textcircled{3} \quad \mathcal{S} = -\frac{1}{2e^2} \int F \wedge *F$$

$$\textcircled{a} \quad \delta \mathcal{S} = -\frac{1}{2e^2} \int \delta F \wedge *F + \frac{1}{2e^2} \int F \wedge \delta(*F)$$

$$= -\frac{1}{e^2} \int \delta F \wedge *F$$

$$= -\frac{1}{e^2} \int d\delta A \wedge *dA = -\frac{1}{e^2} \int \delta A \wedge (d*F) - \frac{1}{e^2} \int d(\delta A \wedge *F)$$

$d*F \equiv \text{E.O.M.}$

$$\delta \mathcal{S} = \int \delta A \wedge \delta A + d\Theta \quad \rightarrow \quad = -\frac{1}{e^2} (*F) \wedge \delta A$$

$$\Theta = -\frac{1}{e^2} \delta A \wedge *F \quad (\text{pre-symplectic form})$$

$$\Omega = \int_{\Sigma} \delta \Theta = \int_{\Sigma} \left(-\frac{1}{e^2} \right) [-\delta A \wedge \delta(*F)]$$

$$= -\frac{1}{e^2} \int_{\Sigma} \delta(*F) \wedge \delta A \quad | \quad \delta \Theta = \omega$$

⑥ $\Sigma = \mathbb{R}^+$, derive the commutators.

on Σ we want $(\delta\phi)u\bar{z}\bar{z}$

$$\leadsto \delta(*F)_{\mu\nu} \wedge \delta A_\sigma dx^\mu \wedge dx^\nu \wedge dx^\sigma$$

$$= \delta(*F)_{uz} \delta A_{\bar{z}} dz \wedge d\bar{z} \wedge d\bar{z} + \delta(*F)_{z\bar{z}} \delta A_z dz \wedge d\bar{z} \wedge dz$$

$$+ \delta(*F)_{z\bar{z}} \delta A_{\bar{z}} dz \wedge d\bar{z} \wedge dz$$

$$\leadsto \delta(*F)_{uz} \delta A_{\bar{z}} - \delta(*F)_{z\bar{z}} \delta A_z + \delta(*F)_{z\bar{z}} \delta A_{\bar{z}}$$

$$\hookrightarrow \delta\phi$$

Note $\odot$ is the same as $\lambda = p_i dq^i$ on $T^*\mathcal{M}$

$$*F)_{\mu\nu} = \frac{1}{2} \epsilon_{\mu\nu\rho\sigma} F^{\rho\sigma}$$

$$(*F)_{uz} = \epsilon_{uzr\bar{z}} F^{r\bar{z}} = -F^{r\bar{z}} = -g^{\bar{z}z} g^{r\alpha} F_{\alpha z}$$

$$= \frac{1}{r^2} \gamma^{z\bar{z}} (F_{rz} - F_{\bar{z}z})$$

$$= \frac{1}{r^2} \gamma^{z\bar{z}} (F_{uz} - F_{\bar{z}z})$$

$$\epsilon_{t123} = \frac{\partial y^\mu}{\partial t} \frac{\partial y^\nu}{\partial x^1} \frac{\partial y^\sigma}{\partial x^2} \frac{\partial y^\rho}{\partial x^3} \epsilon_{\mu\nu\sigma\rho}$$

$$\frac{\partial y^\nu}{\partial x^1} \frac{\partial y^\sigma}{\partial x^2} \frac{\partial y^\rho}{\partial x^3} \epsilon_{uz\nu\sigma\rho}$$

$$= \frac{\partial r}{\partial x^1} \frac{\partial y^\sigma}{\partial x^2} \frac{\partial y^\rho}{\partial x^3} \epsilon_{uz\sigma\rho} + \frac{\partial y^\nu}{\partial x^1} \frac{\partial r}{\partial x^2} \frac{\partial y^\rho}{\partial x^3} \epsilon_{uz\nu\rho}$$

$$+ (\partial y^\nu / \partial x^1) (\partial y^\sigma / \partial x^2) (\partial r / \partial x^3) \epsilon_{uz\nu\sigma r} + \dots$$

$$r^2 = x_i^2 + \dots$$

$$r \partial r / \partial x_i = x_i \Rightarrow \frac{\partial r}{\partial x_i} = \frac{x_i}{r}$$

$$= \frac{1}{r} \left[\left(\frac{\partial y^\sigma}{\partial x^2} \frac{\partial y^\rho}{\partial x^3} \right) x_1 \epsilon_{uz\sigma\rho} + \dots \right]$$

$$= \frac{1}{r} \left[\frac{\partial z}{\partial x^2} \frac{\partial \bar{z}}{\partial x^3} \left(\frac{r(z+\bar{z})}{1+z\bar{z}} \right) \epsilon_{uzr z\bar{z}} + \dots \right]$$

$$= \frac{1}{r} \left[\left(\frac{1}{x^3+r} \right) \frac{(x^1 - ix^2)(-1)}{(x^3+r)^2} \epsilon_{uzr z\bar{z}} \left(\frac{r(z+\bar{z})}{1+z\bar{z}} \right) + \dots \right]$$

$$= \frac{(-i)}{(x^3+r)^2} \frac{\bar{z}(z+\bar{z})}{(1+z\bar{z})} \epsilon_{uzr z\bar{z}}$$

$$(x^3+r) = r \left[\frac{1-z\bar{z}}{1+z\bar{z}} + \frac{1+z\bar{z}}{1+z\bar{z}} \right] = r \frac{2}{1+z\bar{z}}$$

$$\hookrightarrow \frac{(-i) \bar{z}(z+\bar{z})(1+z\bar{z}) \epsilon_{uzr z\bar{z}}}{4r^2}$$

$$\epsilon_{uzr z\bar{z}} = i r^2 \gamma_{z\bar{z}}$$

$$\Rightarrow (\star F)_{u\bar{z}} = \epsilon_{u\bar{z}r\bar{z}} F^{r\bar{z}} = -(\epsilon^{r\bar{z}} \gamma_{z\bar{z}}) g^{\bar{z}z} (g^{rr} F_{rz} + g^{r\bar{u}} F_{\bar{u}z})$$

$$= \underline{i(F_{u\bar{z}} - F_{rz})}$$

$$(\star F)_{z\bar{z}} = \frac{1}{2} \epsilon_{z\bar{z}\mu\nu} F^{\mu\nu} = \frac{1}{2} (\epsilon_{z\bar{z}u\bar{v}} F^{u\bar{v}} + \epsilon_{z\bar{z}r\bar{v}} F^{r\bar{v}})$$

$$= \frac{1}{2} (\epsilon_{z\bar{z}u\bar{r}} F^{u\bar{r}} + \epsilon_{z\bar{z}r\bar{u}} F^{r\bar{u}})$$

$$= \epsilon_{z\bar{z}u\bar{r}} F^{u\bar{r}} = \epsilon_{ur\bar{z}z} g^{u\alpha} g^{\bar{r}\beta} F_{\alpha\beta} = i\epsilon^{r\bar{z}} \gamma_{z\bar{z}} (-1)(-1) F_{ru}$$

$$= \underline{i\epsilon^{r\bar{z}} \gamma_{z\bar{z}} F_{ru}}$$

$$\star F_{u\bar{z}} = i(F_{r\bar{z}} - F_{u\bar{z}})$$

$$\delta\Theta|_{\mathcal{I}^+} = -\frac{1}{e^2} \left\{ \delta[i(F_{u\bar{z}} - F_{rz})] \delta A_{\bar{z}} - \delta[i(F_{r\bar{z}} - F_{u\bar{z}})] \delta A_z \right.$$

$$\left. + i\epsilon^{r\bar{z}} \gamma_{z\bar{z}} \delta F_{ru} \delta A_{u\bar{z}} \right|_{\mathcal{I}^+}$$

$$F_{ru} \sim \mathcal{O}(r^{-1}) \xrightarrow{\mathcal{I}^+} 0$$

$$\partial_r A_z \xrightarrow{\mathcal{I}^+} 0 \quad \text{so} \quad F_{r\bar{z}} \text{ terms vanish as well}$$

$$\delta\Theta|_{\mathcal{I}^+} = -\frac{1}{e^2} \left\{ i \delta F_{u\bar{z}} \wedge \delta A_{\bar{z}} + i \delta F_{u\bar{z}} \wedge \delta A_z \right\}$$

$$= -\frac{i}{e^2} \left\{ \right\} \leftarrow r\text{-independent factor, so } \sim \mathcal{O}(r^0)$$

$$= -\frac{i}{e^2} \left\{ \delta F_{u\bar{z}}^{(0)} \wedge \delta A_{\bar{z}}^{(0)} + \delta F_{u\bar{z}} \wedge \delta A_z^{(0)} \right\}$$

$$\Omega = \int_{\mathcal{M}} i \delta\Theta$$

$$\Omega_{\mathcal{I}^+} = \frac{i}{e^2} \int_{\mathcal{I}^+} d^2z \left[\delta F_{u\bar{z}}^{(0)} \wedge \delta A_{\bar{z}}^{(0)} + \delta F_{u\bar{z}} \wedge \delta A_z^{(0)} \right]$$

$$F_{z\bar{z}}|_{\mathcal{I}^+} \stackrel{!}{=} 0 \quad (\text{No magnetic monopoles})$$

$$\bullet A_z^{(0)} = \hat{A}_z(u, z, \bar{z}) + \partial_z \phi(z, \bar{z})$$

$$= \frac{1}{e^2} \int d^2u d^2z \left[\delta \partial_u (\hat{A} + \partial_z \phi(z, \bar{z})) \wedge (\delta \hat{A}_{\bar{z}} + \partial_{\bar{z}} \delta \phi) \right.$$

$$\left. + \dots \right]$$

$$= \frac{1}{e^2} \int d^2u d^2z \left(\partial_u \delta \hat{A}_z \wedge \delta \hat{A}_{\bar{z}} + \partial_u \delta \hat{A}_z \wedge \partial_{\bar{z}} \delta \phi \right.$$

$$\left. + \partial_u \delta \hat{A}_{\bar{z}} \wedge \delta \hat{A}_z + \partial_u \delta \hat{A}_{\bar{z}} \wedge \partial_z \delta \phi \right)$$

$$= \frac{2}{e^2} \int d^2u d^2z \left(\partial_u \delta \hat{A}_z \wedge \delta \hat{A}_{\bar{z}} + \delta F_{u\bar{z}}^{(0)} \wedge \partial_{\bar{z}} \delta \phi \right.$$

$$\left. + \dots \right)$$

$$= \frac{2}{e^2} \int d^2u d^2z \left(\partial_u \delta \hat{A}_z \wedge \delta \hat{A}_{\bar{z}} + \int d^2z \delta N_z \wedge \partial_{\bar{z}} \delta \phi \right.$$

$$\left. + \int d^2z \delta N_{\bar{z}} \wedge \partial_z \delta \phi \right)$$

$$\text{with } N_z = \partial_{\bar{z}} N$$

$$= \frac{2}{e^2} \int d^2z \partial_z \hat{A}_z \wedge \delta \hat{A}_{\bar{z}} + 2 \int d^2z \delta N_{\bar{z}} \wedge \partial_z (\delta \phi)$$

and so

$$\frac{2}{e^2} \int d^2z \partial_z \hat{A}_z \wedge \delta \hat{A}_{\bar{z}} - 2 \int d^2z \partial_z (\delta \phi) \wedge \delta N_{\bar{z}} = \Omega_{\mathcal{A}}$$

$$\Rightarrow \text{Should have } \Omega = - \int \delta p^i \wedge \delta q^i$$

$$\hookrightarrow [p^i, q^i] = i \delta(\dots)$$

$$\therefore \frac{2}{e^2} [\partial_z \hat{A}_z(z, \bar{z}), \hat{A}_w(w, \bar{w})] = i \delta(z-w) \delta^2(z-w)$$

$$2 [\partial_z \phi(z, \bar{z}), \partial_{\bar{w}} N(w, \bar{w})] = i \delta^2(z-w)$$

integrate...

$$(4) \Delta_\mu(x) = e_{\alpha=-1} \int \frac{d^3q}{(2\pi)^3} \frac{1}{2\omega_q} \left[\xi_\mu^\alpha(\vec{q}) a_\alpha(\vec{q}) e^{i q x} + \xi_\mu^\alpha(\vec{q}) a_\alpha^\dagger(\vec{q}) e^{-i q x} \right]$$

, $\omega = r$

$$\leadsto e^2 \partial_z N = \int_{-\infty}^{\infty} du F_{uz}$$

$$q^1, q^2, q^3 \rightarrow (\omega, \theta, \phi)$$

$$(i) \int \frac{d^3q}{(2\pi)^3} = \frac{1}{(2\pi)^3} \int \int \int d\omega d\theta d\phi \omega^2 \sin(\theta)$$

$$A_\mu(x) = e_{\alpha=-1} \left(\int d\omega d\theta d\phi \frac{1}{(2\pi)^3} \omega^2 \sin(\theta) \left[\xi_\mu^{\alpha} a_\alpha(\vec{q}) e^{i q x} + \xi_\mu^{\alpha} a_\alpha^\dagger(\vec{q}) e^{-i q x} \right] \right)$$

$$(ii) i q x, \quad q^\mu = (\omega, \omega \hat{q})$$

$$x = (u, r \hat{x})$$

$$i q x = -i \omega u + r \omega \hat{q} \hat{x}$$

